The question

Can we rebuild the contacts-v1 training corpora with a decontamination pass that actually covers the eval set we report on — and what does it cost?

Issue #213 measured the overlap: 323 of the 554 eval proteins (58%) have a significant training homolog. This is the other half — the fix, and its price.

What is filtered today

Two corpora, two stories.

AFDB (4,129,682 documents, #53) has NO eval decontamination at all. Its split is afdb-24M's own cluster hash, so nothing leaks across the AFDB split, but the 554 eval proteins were never purged from it. #41 found 99 of 100 FoldBench monomers fall in a fold the model trained on; those verdicts became reporting strata and nothing else.

ESM-Atlas (66,759,922 documents, #91 then #139) has one filter, with four gaps. It dropped 41,517 sequences at 40% identity or above, but: FoldBench-100 was never in the reference, which is 18% of the benchmark; 40% is looser than our own 30% bar; the filter is sequence-only, on the axis #41 showed is the wrong one; and #91's own stated follow-up never happened.

The design

Both corpora are already materialised, so decontamination is a row filter on entry_id, not a regeneration. Three tiers, priced before anything is retrained:

A — sequence. Identity at or above 30% over at least half the eval protein, or E at or below 1e-3. B — plus structurally redundant. A, plus TM 0.90 or above to any eval structure. C — plus fold-level purge. B, plus whole clusters at TM 0.50 or above.

The expensive inputs already existed: #213's 70.9M-sequence MMseqs2 database, and #41's Foldseek database of the 1.33M AFDB cluster representatives. No new pyconfind, no document rebuild, no cluster job.

What we found

Tier A is cheap: 1.89% of AFDB, 1.57% of ESM-Atlas. And it is not an artefact of a threshold — sweeping the search's reporting depth over six decades moves it by a quarter of a percentage point.

All four of #91's gaps are real. Of the surviving contaminated ESM-Atlas rows, 47% sit in the 30-40% band, 24% are remote homologs below any identity bar, 17% are reachable only from FoldBench-100 — and 12% (123,713 rows) clear #91's own 40%/50% rule. That last group is the measured price of its -s 4.0 search sensitivity, a trade-off its own script flagged in a comment.

#41 was right about the axis, and acting on it is cheap at Tier B: 22,320 AFDB documents are structurally near-identical to an eval structure while being invisible to any sequence search. That is 0.54% of the corpus.

Tier C is unaffordable: 37.31% of AFDB, and 95% of that is structure-only. The reason is not an unlucky threshold. The mode of "best TM to any of the 554" sits essentially at the 0.5 same-fold boundary — a third of AFDB's structural clusters simply are the same fold as something in a 554-protein eval set.

Three quarters of that cost protects the 396 de novo designs. Scoping the fold purge to the 158 natural eval proteins costs 9.38% instead of 36.77%. Designs are small idealised bundles that share a fold with an enormous share of AFDB, and they are the proteins with no evolutionary relatives to leak through in the first place.

A wider reference, and a symmetric coverage gate

Two variants were priced on top of the tier ladder.

A wider reference: all of FoldBench, not just the 100 monomers we score. Its protein-protein, antibody-antigen, protein-peptide, protein-ligand, protein-DNA and protein-RNA tasks carry protein chains too — 1,940 of them across 1,493 entries, and all 100 scored monomers are inside that set.

A symmetric coverage gate: Tier A gates on coverage of the eval protein, which misses a short training protein aligning to one domain of a long eval protein. Gating on the shorter of the two sequences closes that.

Under "30% identity over at least half of the shorter sequence", with no E-value arm at all, the union of the 554 and all of FoldBench drops 4.04% of AFDB and 1.81% of ESM-Atlas — 1,373,423 of 70,889,604 training proteins, or 1.94% overall.

All of FoldBench costs more AFDB than our own eval set does (3.16% vs 1.39%), which is expected: its chains are all natural PDB proteins with real evolutionary families, where 396 of our 554 are de novo designs with almost nothing to purge. The union is well below the sum, 86% of it in both arms, because the FoldBench monomers sit in both references and a training protein is routinely homologous to several eval proteins at once.

The coverage choice is worth about 1.1 points of AFDB: the shorter-sequence gate drops 4.04% where the reference-side gate drops 2.92%. Adding Tier A's remote-homology arm on top takes it to 5.69% / 3.03%.

Even the widest reference with the most permissive coverage gate leaves 96-98% of the training data intact, against the 37% Tier C alone would delete.

What was published

Both corpora were rebuilt under the rule chosen after seeing these numbers: 30% identity over at least half of the shorter sequence, against the 554 and all of FoldBench, with no E-value arm. Sequence axis only.

contacts_v1_decontam 4,129,682 -> 3,963,003 (166,679 removed, 12.1 GB)
contacts_v1_esm_atlas_decontam 66,759,922 -> 65,553,178 (1,206,744 removed, 130.7 GB)

Both are published as new prefixes; the originals are untouched, so every existing checkpoint stays reproducible against the corpus it actually saw. The rebuild is a row filter and nothing else — same shard numbering, same row order, same parquet codec, byte-identical surviving documents.

Verified on the bucket rather than on the build: reading the entry_id column back off all 2,067 and 3,338 published shards finds no missing shard, exactly the expected row counts, and zero surviving rows from the drop list.

Where it lands

H1 holds through Tier B; H0 holds for Tier C. Tier C is declined, with 37.31% as the number that justifies declining it; the 9.38% natural-only variant is a real middle option if fold novelty later becomes load-bearing.

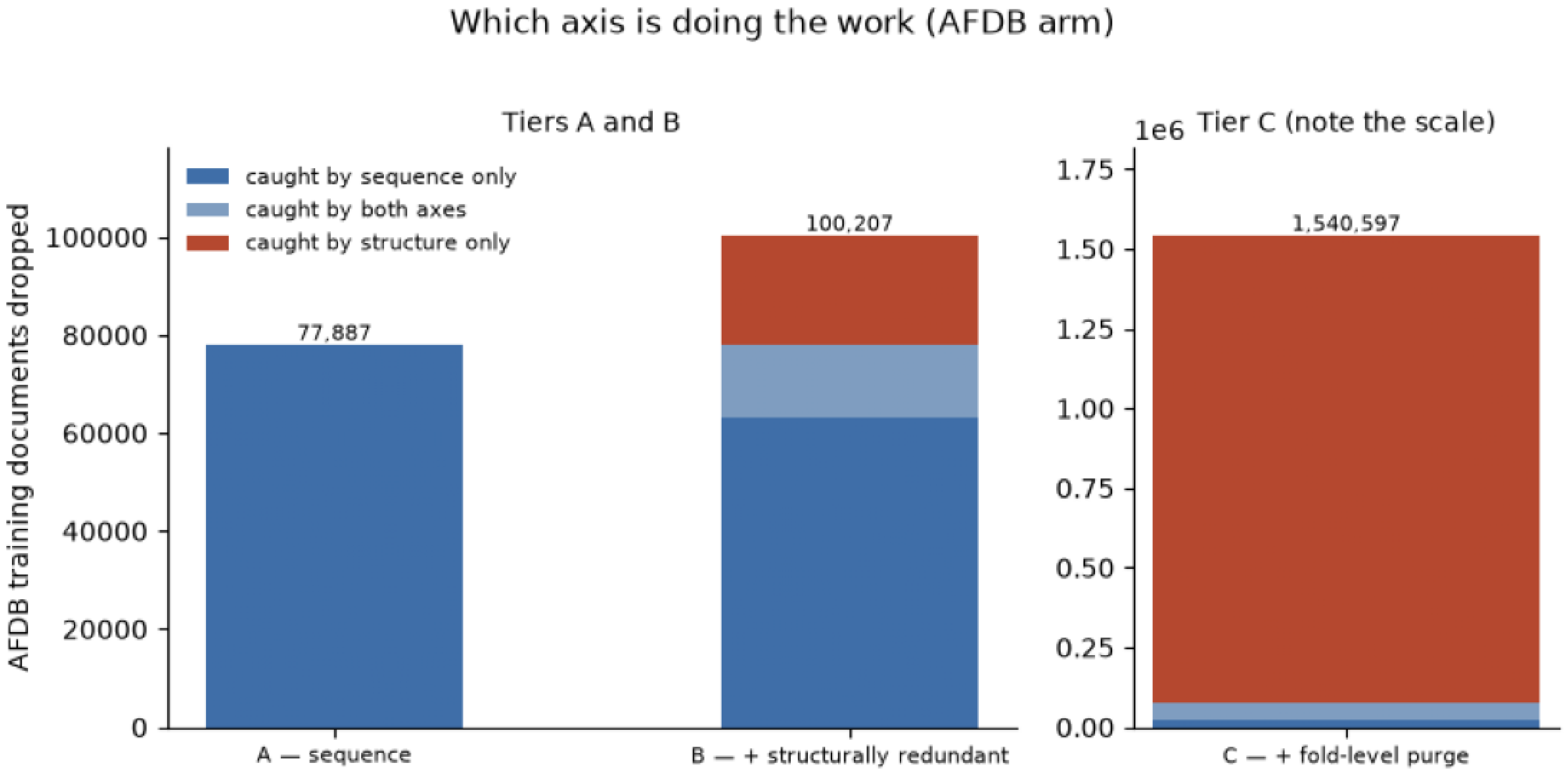
What was actually published is neither Tier A nor B but the rule above — sequence-only, but against a wider reference (the 554 plus all of FoldBench) and with the symmetric coverage gate. It costs 4.04% of AFDB and 1.81% of ESM-Atlas, more than Tier A's 1.89%/1.57% and on a broader reference, and it leaves the structural axis unapplied.

Do not pay for the ESM-Atlas Foldseek build (about $1k) yet. It was gated on this table. The only tier it could serve is B, since C is declined, and on the arm we can measure, B's structural increment is 0.54%. Revisit if the retrain moves. The caveat: ESM-Atlas is metagenomic, so its fold distribution against this eval set need not match AFDB's.

Stage 5 — retraining the #199 recipe on the decontaminated mixture and re-scoring against #213's homology-free subset — is the actual test of H1. Nothing here measures accuracy.

axis_decomposition.png

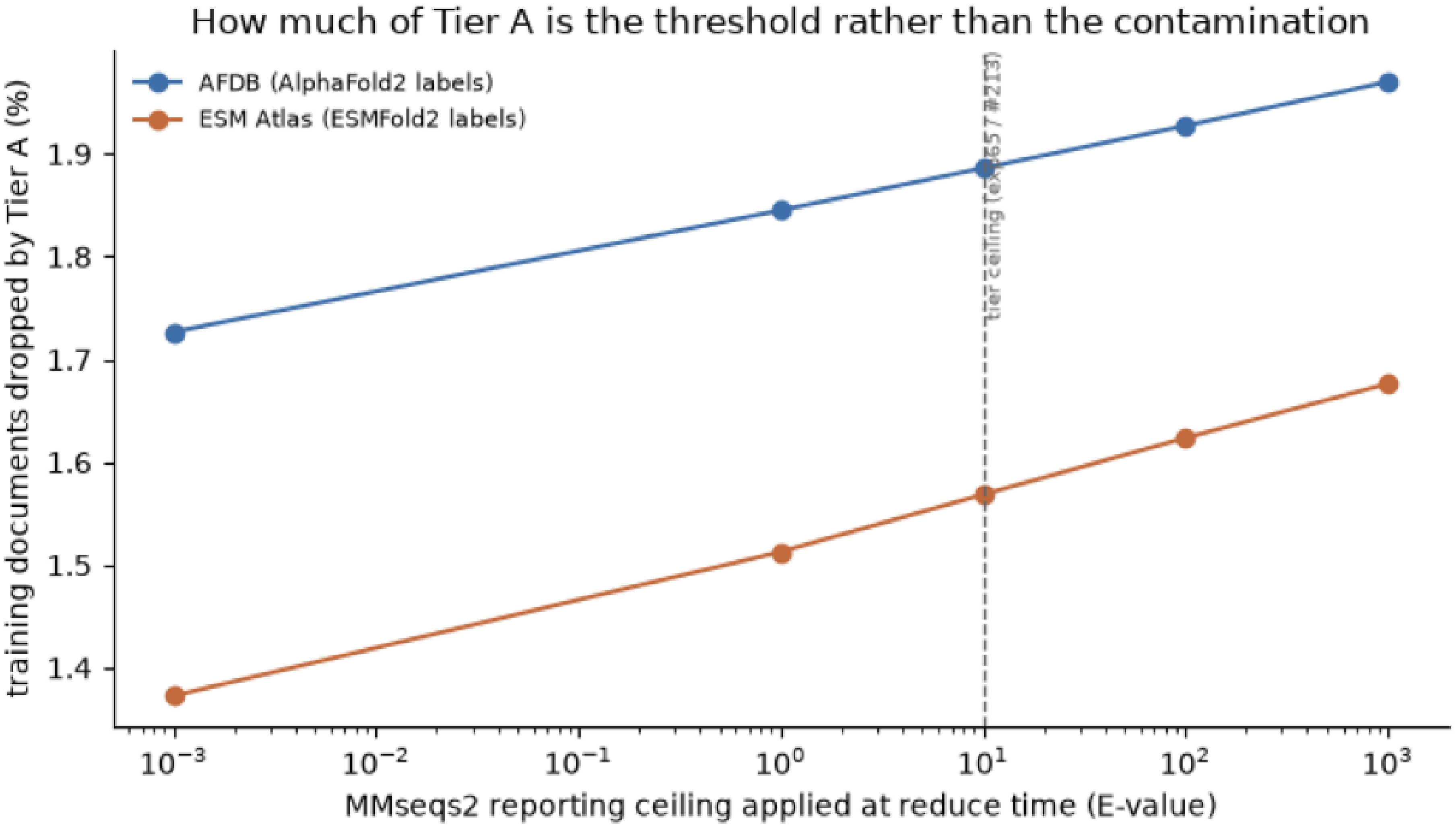
Red = what #41 predicted and nobody had measured — documents a sequence filter keeps and a structural one removes: 22,320 at Tier B, 1,462,710 at C.



`$ plot_decontam.py`

evalue_sensitivity.png

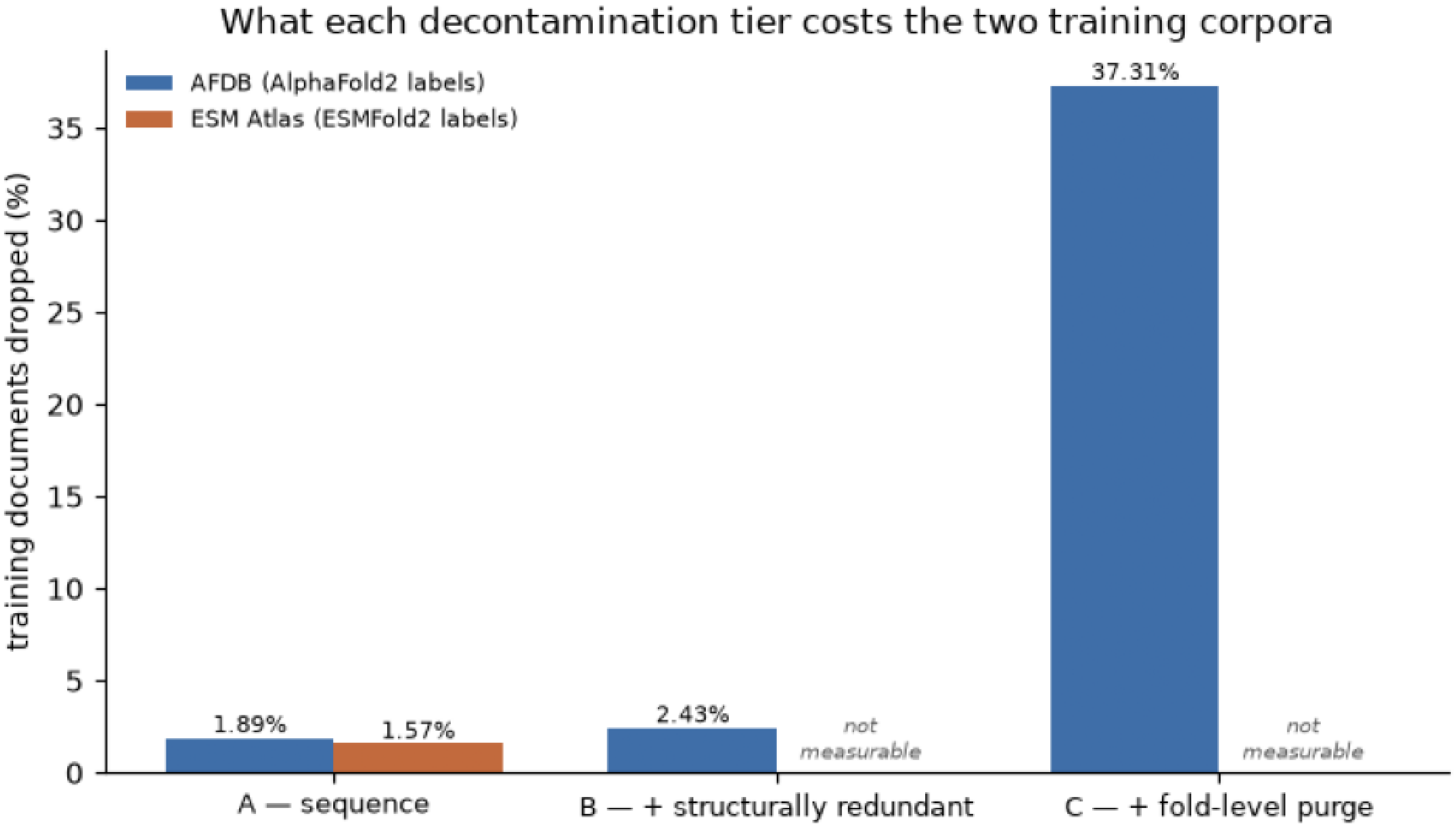
The E-value arm of Tier A contributes nothing above 1e-3; everything the curve gains after that is the identity arm, which has no significance floor of its own.



$ plot_decontam.py

survival_by_tier.png

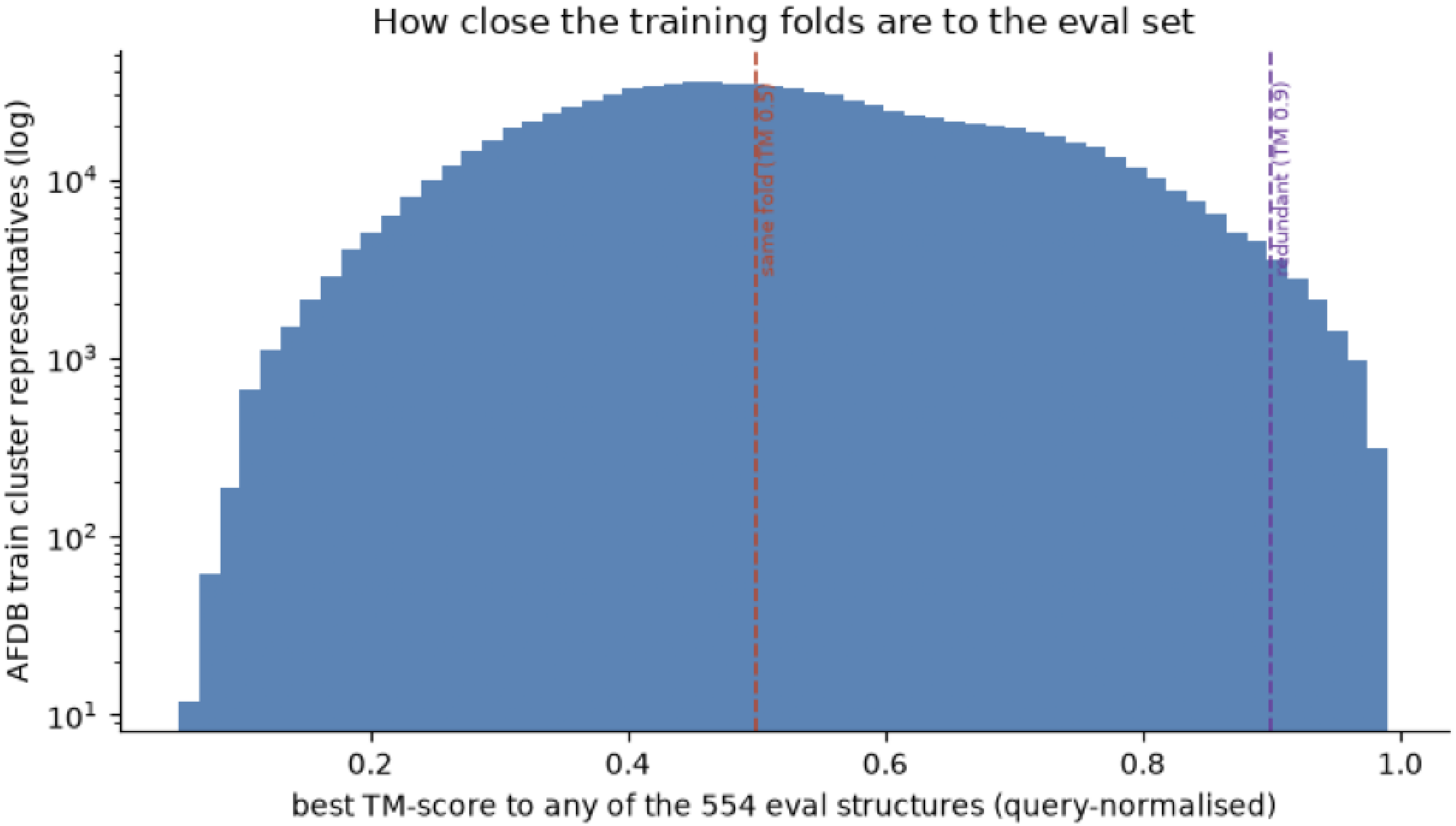
Share of each published train corpus removed at each tier. ESM-Atlas has no structural database, so its Tier B/C cells are unmeasured rather than zero — building one is the job this table gates.



$ plot_decontam.py

tm_distribution.png

945,861 of the 1,304,911 AFDB train representatives got a hit and are counted here; the other ~359k had none and are off the left edge.



`$ plot_decontam.py`